
Computational Lithography Solutions for Non-Immersion ArF (193nm) Excimer Laser Deep Ultraviolet Machines for Chemically Amplified Resists with Bottom Anti-Reflective Coating

Executive Program in Semiconductor Manufacturing -
IIT Delhi

Tejas Beedkar
Sushma Shetty
Nitesh Mahawar
Juganta Mosrong

Index

Chapter 1 - Introduction

Chapter 2 - Abbreviations and Concepts

- 2.1 Abbreviations
- 2.2 Diffraction
- 2.3 Chromatic Aberration
- 2.4 Spatial Coherence
- 2.5 Laser Speckle
- 2.6 Full Width at Half Maximum (FWHM)
 - 2.6.1 The Mathematical Framework Behind FWHM
 - 2.6.2 Technical Significance in DUV Lithography

Chapter 3 - ArF Laser Cabinet

- 3.1 ArF Excimer Laser Cabinet
 - 3.1.1 — Beam Conditioning Chain (Raw Excimer Output → Illuminator)

- 3.1.2 Key Beam Characteristics: Source vs. Illuminator-Stage Delivery with Component → Parameter Mapping
- 3.2 Excimer Spectrum (Raw Cavity Output and Ideal)
 - 3.2.1 Practical Excimer Spectrum (Raw Cavity Output)
 - 3.2.2 Ideal Excimer Spectrum for DUV Photolithography
- 3.3 Line Narrowing Module (LNM) / Line Narrowing Package (LNP)
 - 3.3.1 Beam Expander (Prism Pair)
 - 3.3.2 Echelle / Littrow Diffraction Grating
 - 3.3.3 Etalon (Fabry-Pérot)
- 3.4 Metrology - Wavelength Module
- 3.5 Beam Conditioning & Delivery Unit (BDU)
 - 3.5.1 Variable Attenuator
 - 3.5.2 — Beam Expanding Telescope
 - 3.5.3 — Coherence Scrambler / Light Pipe
 - 3.5.4 — DUV Bandpass Filter (KrF systems)

Chapter 4 - Illuminator

- 4.1 Step 1 - Homogenisation
 - 4.1.1 Step 1a - First Fly's Eye Lens Array (Beam Homogenizer #1)
 - 4.1.2 Step 1b - Second Fly's Eye Lens Array (Beam Homogenizer #2)
- 4.2 Step 2 - Pupil Shaping
 - 4.2.1 Step 2a - Diffractive Optical Element (DOE)
 - 4.2.1.1 Function of the DOE
 - 4.2.1.2 DOE matching
 - 4.2.1.3 Diffractive Optical Element (DOE) Types — Pattern Suitability
 - 4.2.2 Step 2b - Partial coherence factor (σ)
 - 4.2.2.1 σ trade-off physics
 - 4.2.2.2 What $\sigma=0$ breaks
 - 4.2.2.3 Outer σ control iris
 - 4.2.3 Mixed Pattern Pupil Shaping
 - 4.2.3.1 Mixed Layer Compensation toolset
 - 4.2.3.2 Evolution by technology node
- 4.3 Step 3 - Relay Lens * 4.2.1 Step 3a - Condenser / Relay Lens Group * 4.2.2 Step 3b - REMA Masking Blades * 4.2.3 Step 3c - REMA relay lens**
- 4.4 Stage 4 - Dose Control

Chapter 5 - Reticle Design in Lithography

- 5.1 Introduction
- 5.2 Reticle Design, Evolution, and AI Integration
 - 5.2.1 Historical Evolution of Reticle Design
 - 5.2.2 Evolution and Mechanics of Modern Reticle Design
 - 5.2.3 Computational Lithography: The Foundation

- 5.2.4 Software: EDA Software and Data Preparation
 - 5.2.5 The Historical and AI Evolution
- 5.3 Advanced Reticle Design and the AI Revolution
 - 5.3.1 The 4x Paradigm and EUV
 - 5.3.2 Types of Reticle Patterns
 - 5.3.3 The AI Revolution
 - 5.3.4 Future Directions: Beyond Monolithic Reticles
- 5.4 The 193nm Epoch: Pushing the Limits of Optical Lithography
 - 5.4.1 Historical Evolution and the 193nm Era
 - 5.4.2 The Rayleigh Criterion
 - 5.4.3 The Shift to Immersion
 - 5.4.4 Resolution Enhancement Techniques (RET)
 - 5.4.5 Economic Context
 - 5.4.6 The 193nm Immersion (193i) Workhorse
- 5.5 Manufacturing-Design Interface
- 5.6 Visualizing Reticle Patterns
- 5.7 Challenges faced by AI-optimized Reticles
- 5.8 Conclusion

Chapter 6: Projection Lens in Reduction Photolithography

- 6.1 Role of the Projection Lens in Reduction Lithography
- 6.2 Computational Lithography and Mask Optimization
- 6.3 Resist Process Co-Optimization and Depth of Focus
- 6.4 Illumination and Recipe Selection Around the Lens
- 6.5 Practical Fine-Tuning of Projection Lenses in Fabs
- 6.6 Primary Failure Modes with Respect to the Projection Lens
 - 6.6.1 Fundamental Resolution / NA Limit
 - 6.6.2 Thermal Distortion and Drift of the Lens
 - 6.6.3 Residual and Evolving Aberrations

Chapter 7 Wafer and Photoresist Stack

- 7.1 Introduction
- 7.2 Silicon Wafer Substrate
 - 7.2.1 Wafer Foundation
 - 7.2.2 Importance of Wafer Flatness
 - 7.2.3 Optical Interaction with Silicon
- 7.3 Need for Anti-Reflective Coatings
 - 7.3.1 Reflection Mechanism
 - 7.3.2 Standing Wave Formation
 - 7.3.3 Swing Curve Effect

- 7.4 Bottom Anti-Reflective Coating (BARC)
 - 7.4.1 Purpose
 - 7.4.2 Reflection Cancellation Principle
 - 7.4.3 BARC Modeling in Computational Lithography
 - 7.4.4 Impact on Lithographic Performance
- 7.5 Top Anti-Reflective Coating (TARC)
 - 7.5.1 Purpose
 - 7.5.2 Reflection Cancellation Principle
 - 7.5.3 Benefits
 - 7.5.4 TARC Modeling in Computational Lithography
- 7.6 Chemically Amplified Resist (CAR)
 - 7.6.1 Purpose
 - 7.6.2 Resist Composition
 - 7.6.3 Exposure Process
 - 7.6.3.1 Photoacid Generation Model
 - 7.6.4 Latent Image Formation
 - 7.6.4.1 Optical Absorption within the Resist
 - 7.6.5 Post-Exposure Bake (PEB)
 - 7.6.5.1 Acid Diffusion Model
 - 7.6.6 Development
 - 7.6.6.1 Threshold Development Model
 - 7.6.7 CAR Modeling in Computational Lithography
 - 7.6.8 Impact on Lithographic Performance
- 7.7 Integrated Wafer Stack and Pattern Formation
 - 7.7.1 Introduction
 - 7.7.2 Complete Wafer Stack
 - 7.7.3 Pattern Formation Mechanism
 - 7.7.4 Sources of Pattern Degradation
 - 7.7.5 Relationship Between Wafer Stack and Lithographic Performance
- 7.8 Chapter Summary

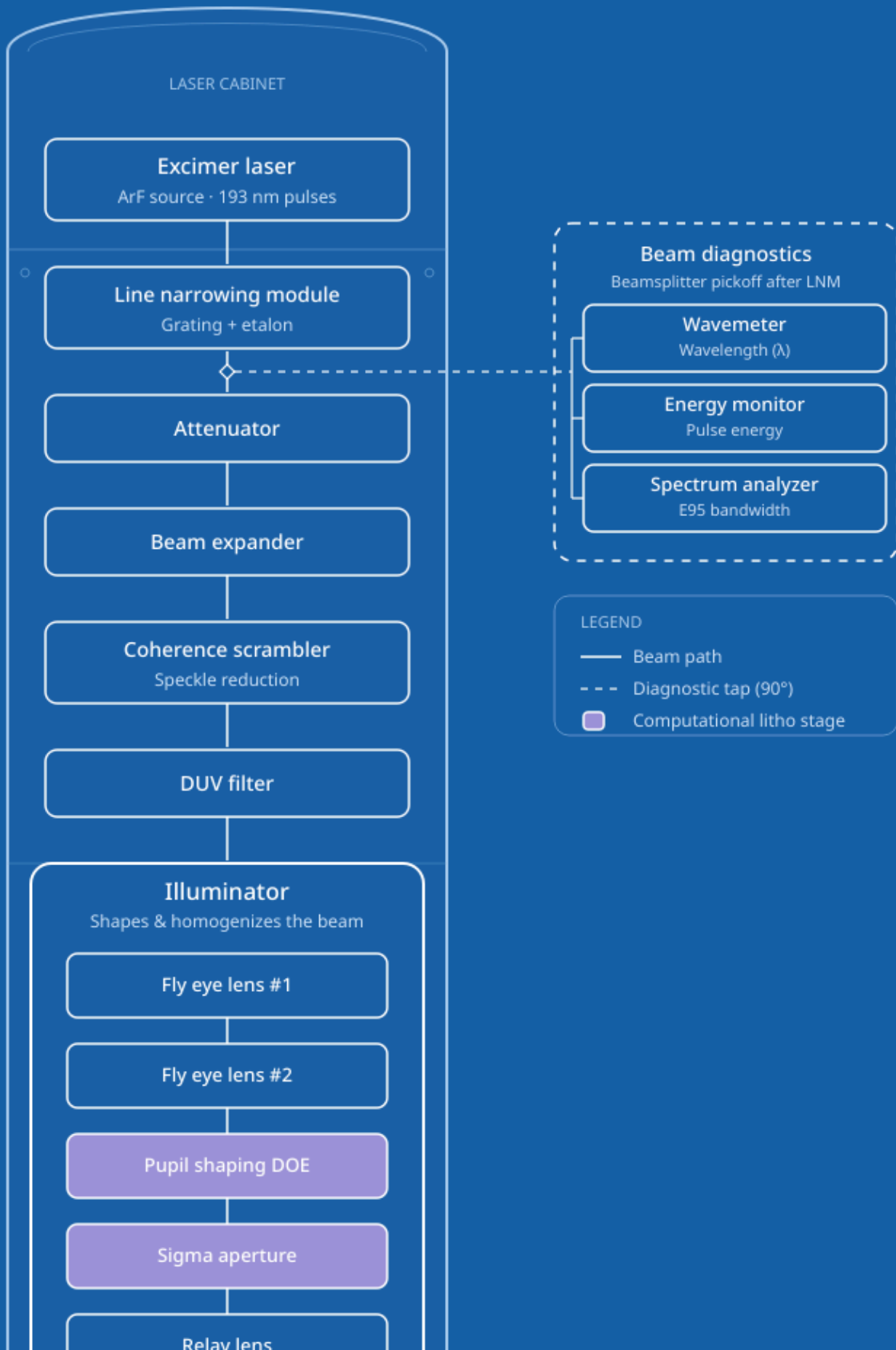
Chapter 1 - Introduction

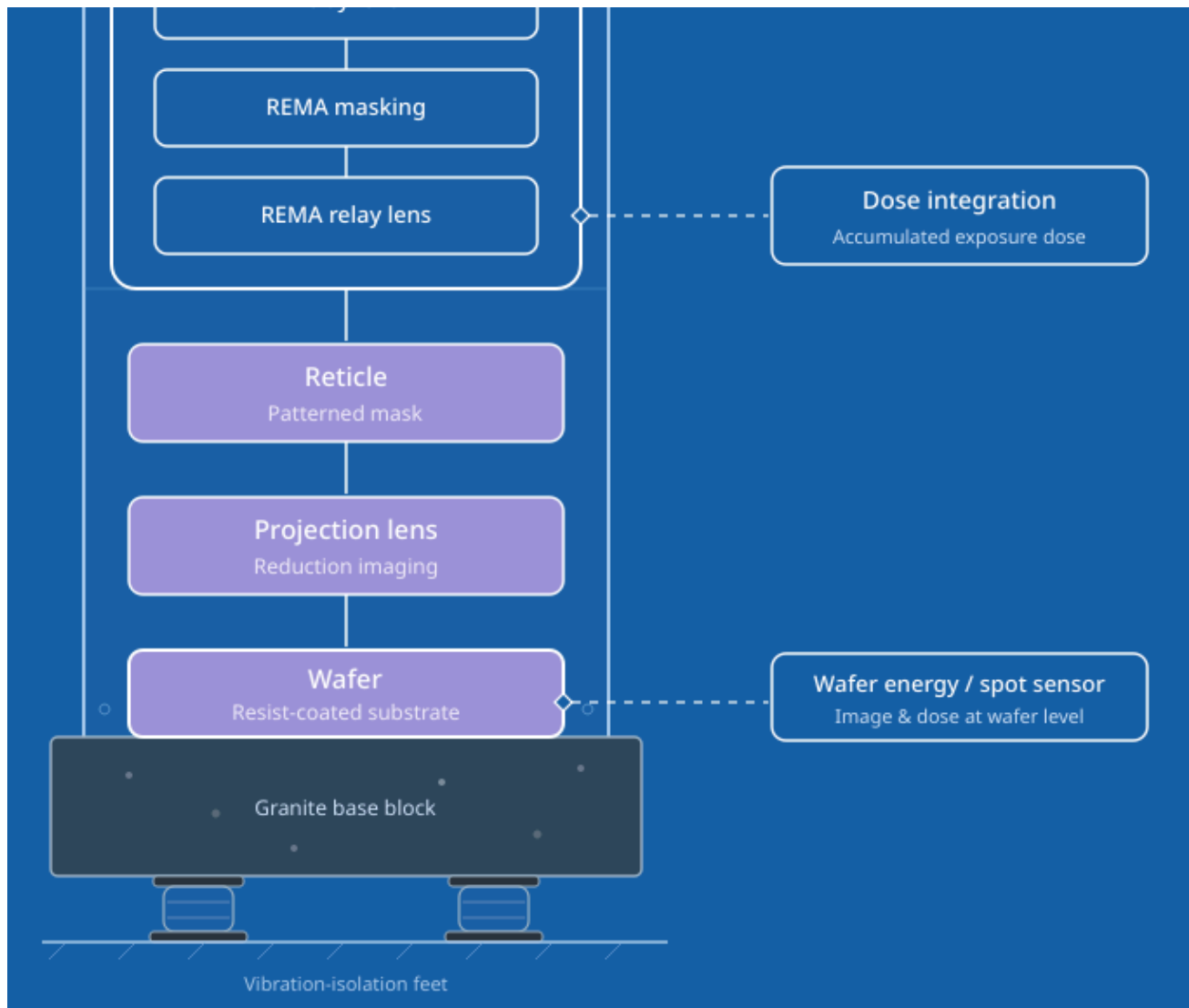
Argon-fluoride (ArF) non-immersion lithography, often called "dry" 193 nm DUV, is a deep-ultraviolet projection system. It prints integrated-circuit patterns onto a resist-coated wafer using a 193 nm excimer laser. The technology occupies a specific rung in the long ladder of optical lithography. The earliest steppers used mercury-lamp lines: g-line at 436 nm and i-line at 365 nm. The industry then moved to deep-ultraviolet excimer sources, first KrF at 248 nm and then ArF at 193 nm. Dry ArF was the workhorse that carried the industry through roughly the 90 nm to 45 nm

nodes. It was later extended by ArF immersion (193i), which fills the gap between the final lens and the wafer with water to raise the effective numerical aperture. At the smallest nodes it was ultimately succeeded by EUV at 13.5 nm. The "dry" qualifier simply distinguishes it from that immersion variant: the final imaging space is air, which caps the numerical aperture below 1.0.

Dry ArF non-immersion DUV pipeline

Beam path runs vertically · metrology branches 90° off-axis (dashed)





In operation, the excimer laser fires 193 nm pulses. These are first spectrally purified by a line-narrowing module, which uses a grating and etalon. The beam is continuously checked by beam diagnostics — a wavemeter, energy monitor, and spectrum analyzer — that sample it through a pickoff. The conditioned beam is then attenuated, expanded, and de-cohered. It passes through a DUV filter into the illuminator. There, a pair of fly-eye lenses homogenize the light. A pupil-shaping diffractive element and sigma aperture define its angular distribution. Relay and REMA optics shape and mask the illumination field, while a dose-integration sensor tracks accumulated exposure.

This carefully formed beam illuminates the reticle, the patterned mask. The projection lens then demagnifies that pattern and images it onto the wafer. The wafer itself is a layered stack: a chemically amplified resist (CAR) sits on top of a bottom anti-reflective coating (BARC). The BARC suppresses reflections from the underlying substrate that would otherwise distort the image, and the CAR uses a photo-generated acid to amplify its response to the weak 193 nm dose. The resist is exposed one field at a time as the stage steps and scans across the substrate.

Within this machine, our computational-lithography solution does not touch every component. It acts on a specific set of elements that together govern the printed image.

These are shown in Violet in the above image.

It produces an optimized illumination shape, which is physically realized by the pupil-shaping DOE and sigma aperture and an optimized mask pattern on the reticle, including optical-proximity corrections and sub-resolution assist features. In the most advanced source-mask-pupil flows, it also makes fine wavefront adjustments at the projection lens.

The wafer is not configured at all. It is the target — the surface whose final printed result the optimization is solving for. This layer is needed because 193 nm light is being asked to print features far smaller than its own wavelength. In that regime, diffraction and optical proximity effects severely distort the image, and a naïve mask no longer prints faithfully. Computational lithography co-optimizes the source and the mask, and sometimes the lens. The goal is to make the light interfere in just the right way to reconstruct the intended geometry on the wafer, despite the wavelength limit.

This makes Computational Lithography the essential bridge between a fixed-wavelength optical system and the ever-shrinking patterns it must resolve.

References

https://www.researchgate.net/publication/258762232_Effects_of_prism_beam_expander_and_slits_on_

https://www.researchgate.net/publication/253151871_High-repetition-rate_ArF_excimer_laser_for_193-nm_lithography

<https://www.spiedigitallibrary.org/conference-proceedings-of-spie/12496/124962U/The-development-of-a-machine-learning-based-excimer-laser-performance/10.1117/12.2658043.short>
